

RESEARCH ARTICLE

A Hybrid Deep Learning Architecture for Social Media Bots Detection Based on BiGRU-LSTM and GloVe Word Embedding

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ABSTRACT Social media platforms have opened avenues for communication, information sharing, and engaging with others online. Automated accounts, known as social media bots, have been observed to engage in harmful activities such as disseminating misinformation, participating in online propaganda and election interference, spreading spam, cyberbullying, and harassing people. This paper proposes a new hybrid architecture based on semantic word embedding and Recurrent Neural Networks (RNNs) to detect social media bots. The research methodology includes the use of Global Vectors (GloVe) for text representation to convert tweets into vectors and combining the Bidirectional Gated Recurrent Units (BiGRU) and Long Short-Term Memory (LSTM) algorithms for semantic text-based classification. Using the proposed architecture, the training process was conducted with two datasets, Cresci-2017 and Twibot-20. The effectiveness of the approach in detecting automated accounts was assessed using five evaluation metrics: Precision, Accuracy, Recall, and F1-score. The proposed approach showed outstanding results in identifying social media bots based only on text-based content, achieving a Precision of 100%, Accuracy of 99.73%, Recall of 99.56%, and F1-Score of 99.63% using the Twibot-20 dataset. Moreover, the proposed architecture surpassed the results obtained by the state-of-the-art approach and showed resilience to overfitting and the ability to detect social media bots effectively in unseen and recent data. This highlights the importance of utilizing deep learning methods and semantic word representations to effectively address issues related to detecting and managing social media bot operations.

INDEX TERMS BiGRU, GloVe, hybrid RNN architecture, LSTM, social media bots detection, SMOTE-ENN, stratified k-fold cross-validation.

I. INTRODUCTION

Online Social networks (OSNs) have become popular platforms for internet users. These platforms enable users to connect with employers, friends, and family, create communities and professional networks to grow their online presence and influence, access job opportunities, and stay up-to-date by following breaking news, trending events, and topics [1]. According to official statistics of “statistica.com” for 2023, 4.95 billion social media users represent 61.4% of the world’s population. This indicates that OSNs are popular and used

by more than half of the world’s population. These statistics have shown that 3.03 billion active users use Facebook every month, 2.491 billion potential advertising reach on YouTube, there are at least 2 billion monthly active users on WhatsApp, and 1.504 billion active users active every month on Instagram and WeChat, including 1.24 billion active users every month. However, not all OSN users are humans, several types of automated accounts are active in OSN, these types of users are called social media Bots (SMBs) [2]. SMBs are automated programs designed to perform specific activities on social media that can be malicious or legitimate. Malicious SMBs are used to manipulate public opinion; spread fake news and information; generate artificial engagement and

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inflate likes, comments, and shares on posts; spread spam; manipulate public opinion by amplifying specific narratives and hashtags; create a false sense of consensus; influence democratic processes; and create the illusion of popularity [3]. SMBs have also been part of the increase in hate speech, harassment, and trolling, which is equally concerning as they may be deployed to inflict psychological harm on individuals or groups through targeted abuse, threats, and coordinated attacks [4]. The presence of SMBs has raised some critical questions about whether information on social media is authentic and reliable. As a result, policies and measures are implemented in OSNs to detect and combat the spread of malicious SMB. However, the sophisticated nature of contemporary bots presents significant challenges to their detection. Their ability to mimic human behavior, adapt to detection algorithms, and operate within complex networks requires the development of advanced detection systems that can tackle SMB activities. Despite improvements in AI-driven social bot detection techniques, especially owing to advances in machine learning and deep learning, the emergence of novel bots remains critical. This highlights the need for continuous research and enhancement of detection systems [5]. A previous study [4] showed that OSN content attributes were the most discriminant features for SMB detection. Content-based detection approaches analyze the content posted by users, such as posts, tweets, messages, or comments, to determine if it is likely to be created by humans or bots. These methods offer several advantages, including a deeper understanding of context, linguistic understanding, and sentiment as well as uncovering hidden connections within tweets and comments [6]. Deep-learning methods require text to be transformed into numerical data. For this purpose, two methods are commonly used: statistical embedding and text embedding. Even if statistical approaches such as Latent Dirichlet Allocation (LDA) and term frequency-inverse document frequency (TF-IDF) are interesting, they can miss subtle relationships between data, especially those involving complex semantics, context, and nuanced language use. Compared to statistical-based text analysis, word embedding is considered a cornerstone of advanced natural language processing (NLP). They allow the extraction of features from the text by considering semantic relationships between words, enhancing generalizability to unseen data, eliminating the need for extensive feature engineering, and integrating effectively with deep learning models [7]. Recurrent neural networks (RNNs) are deep learning algorithms that excel in processing sequential data such as text, understanding context, extracting semantic relationships, identifying key topics, and analyzing sentiments and emotions within text-based content.

To efficiently detect SMBs based on comments, this study proposes a new hybrid deep learning architecture that leverages the combined strength of word embedding and RNN algorithms. The GloVe method converts tweets into semantic vectors for textual representation. These vectors then serve as inputs for a hybrid RNN architecture that combines

bidirectional gated recurrent units (BiGRUs) and Long Short-Term Memory (LSTM) networks. This combination enables the model to perform semantic and context-based classification of user accounts for efficient social media bot detection. The primary contributions of this study are as follows:

- This research explores social media bot detection methods, focusing on text-based features and the binary classification of online social network accounts.
- The proposed architecture for social media bot detection uses a combination of GloVe embeddings, BiGRUs, and LSTM networks, which makes it a unique and novel contribution to the field of social bot detection.
- This approach leverages a semantic text representation achieved through word-embedding techniques using GloVe-based feature extraction. Notably, a hybrid architecture was employed by combining the BiGRU and LSTM models. This architecture analyzes the bidirectional sequences of all tweets posted by an OSN account, enabling the identification of the characteristics indicative of SMB detection.
- The efficiency of the proposed SMB detection system was rigorously assessed through comprehensive experimentation using the Cresci-2017 and Twibot-20 datasets. A Synthetic Minority Oversampling Technique and Edited Nearest Neighbors technique was employed to address potential class imbalance issues and ensure balanced data representation during training.
- The experimental results were validated using several evaluation metrics, including (precision, accuracy, recall, and f1-score) and Stratified 5-Fold Cross-Validation, which is an advanced version of cross-validation, and tested using recent and unseen data.

The structure of this paper is organized as follows. Section II explores the current political social media bot landscape, examines the approaches used for detection, and compares the state-of-the-art approaches comprehensively. Section III presents the background of the deep learning techniques and word-embedding algorithms. Section IV presents the proposed methodology encompassing all stages of model development and addresses the challenges encountered in SMB detection, such as imbalanced data and accurate model training, testing, and validation. Section V shows the experimental results of this approach. Finally, Section VI presents conclusions and avenues for future research.

II. RELATED WORK

The increasing number of social media bots presents an essential threat to online platforms' security and privacy, affecting discourse manipulation, disinformation campaigns, and spams. The accurate detection of these automated accounts is critical for maintaining the platform integrity of OSN and user trust. This section examines existing social media bot detection research, notable findings, and performance assessments, emphasizing significant methodologies, breakthroughs, and limitations.

A. LITERATURE REVIEW

In this article [8], the authors proposed an SMB detection system based on word embedding and machine learning (ML) algorithms. To build their models, they extracted the Content feature (CF) from the Cresci-2017 dataset. As a result, Using Random Forest (RF) with Bert achieved a precision score of 99.96%, and the DT algorithm combined with Doc2Vec achieved a precision score of 100%. The main limit of this work is the use of ROS can introduce overfitting issues. Additionally, the method's applicability appears limited to spam bot detection. The authors of this article [9] created a spammer SMB detection technique that used the posting relationships between users and messages to combine the detection of social spammers and spam messages. For this purpose, the authors used a dataset of microblogs extracted from the Sina Weibo platform, which included CF and Network Features (NF). They employed a Logistic Regression (LR) algorithm to detect social spammers and spam messages and achieved an accuracy of 95%, a central strength of this method lies in its ability to capture semantic relationships between words; however, employing random oversampling as a technique can introduce overfitting issues. Additionally, the method's applicability appears limited to spam bot detection. DABot [10] is a novel framework devoted to detecting active social media bots on Sina Weibo using CF and deep-learning algorithms. DABot combines two key strategies: advanced feature engineering with active learning for data expansion. Thirty features across four categories were extracted to train the proposed deep neural network, incorporating Resnet, BiGRU, and an attention mechanism called RGA. This approach achieved an accuracy of 98.87%, the key limitation of this work is that it needs semantic and context representation of content. This study [11] proposed a sentiment-based SMB detection system using content analysis. The authors used datasets scraped from Twitter, which included CF, User Information (UI), and sentiment features. The model was built using different ML algorithms and Latent Dirichlet Allocation (LDA) for topic modeling. The best results were achieved using AdaBoost with an F1-score of 93.88%, limited topic interpretation in LDA, and difficulty in assigning clear labels or understanding semantic structure. The authors of this research [12] presented a hybrid system for detecting automated Twitter spammers using Bert and ML algorithms. Using the Cresci-2017 dataset and extracting user, text, network, and temporal features, the best performance was reached by the Feed Forward Neural Network (FFNN) algorithm with an accuracy of 95%, This method leverages the strengths of BERT to capture semantic meaning from textual data, but its current focus on spam bot detection may limit its wider applicability. DeepSBD [13] is an automated solution for detecting SMBs in current Twitter accounts. This proposition is based on a Convolutional Neural Network (CNN), GloVe embedding for text presentation, and bidirectional long short-term memory neural network (BiLSTM) with attention mechanisms. Five Twitter datasets

were employed to build the classification model, comprising user, content, network, and temporal features. The SMOTE method was employed to deal with unbalanced data. The approach identified bots on Twitter with an accuracy of 97% to 99%. The combination of DNNs and attention mechanisms has shown promise in bot classification, but its current application is limited to spam bot detection. This study [14] presented a Twitter SMB detection system using BiLSTM and GloVe word embedding. The authors used the Cresci-2017 dataset, which includes UI, CF, network, and temporal features. The proposed approach achieved an accuracy of 92.9% for the detection of Twitter SMBs it can capture the semantic relationships in CF, but it's only dependent on spambot detection. In this article [15], the authors proposed a detection system based on UI and CF by combining CNN and artificial neural network (ANN) models. The authors used The Cresci-2017 dataset to implement the models, which reached an accuracy above 90%. Therefore, it was limited to understanding the semantic structure of the content and could only detect spam bots. This study [16] focuses on detecting fake news related to COVID-19. The approach used BERT for the feature extraction step, and the FFNN model was trained on the Cresci-2017 dataset. The results indicated an accuracy of 86%, however can only detect spam bots and fake news. The study [17] explored the potential of five machine-learning algorithms to detect fake Instagram users. The Random Forest model achieved an accuracy of 91.76% using UI and CF, the primary limitation of this approach lies in its capacity to solely identify fake accounts and its inability to comprehensively grasp the semantic nuances of the content. This research [18] presented an SMB detection system based on the FFNN algorithm using only user information features from the Cresci-2017 dataset. The combination of the ELMO and GloVe word embedding methods with the FFNN model boosted the detection accuracy with a score of 94%. This approach leveraged the strength of deep contextualization to effectively detect social bots in text-based features. To address the growing challenges of SMBs manipulation on Twitter, especially after the 2016 US elections, this study [19] leveraged the power of neural network ensembles. By combining CNN and LSTM models, the proposed approach analyzes textual content using the Cresci-2017 dataset to classify bot tweets and combat misinformation spread an accuracy of 97%, The key limitation of this strategy appears to be that it is only useful for detecting spam bot. BGSRD combines a Graph Convolutional Network (GCN) and a graph neural network (GNN) to address the rise of misinformation and malicious content produced by SMBs [20]. The approach included the use of BERT representations to capture hidden features. Five datasets were used to build the model: Cresci-Rtbust [21], Botometer-Feedback [22], Gilani-2017 [23], Cresci-Stock-2018 [24], [25], and midterm [26]. As a result, the model achieved an accuracy of 80%, the main strength of this approach is its ability to detect social bots with small content length.

TABLE 1. Comparative table of the state-of-the-art articles.

Reference	Algorithm	Features	Dataset	Balancing	Word Embedding	Hyperparameters Tuning	Results
[9]	Logistic Regression	CF, UI	Weibo	ROS	Word2vec	—	A: 92%
[12]	FFNN, Bert	CF	Cresci-2017	—	GloVe	Grid search	A: 95%
[8]	RF	CF	Cresci-2017	ROS	Various	—	A: 99%
[10]	Restnet, BiGRU, Active learning	CF	Sina Weibo dataset	—	—	—	A: 98.87%
[11]	Adaboost	CF, UI	Twitter dataset	—	LDA	—	F1-score: 93.88%
[13]	BiLSTM, CNN, attention	CF, TF, UI, NF, BF	Cresci-2017	SMOTE	GloVe	—	A: 97%
[14]	BiLSTM	CF, TF, UI, NF, BF	Cresci-2017	ROS	GloVe	—	A: 92.9%
[15]	CNN and ANN	CF, UI	Cresci-2017	—	—	—	A>90%
[16]	FFNN	CF	Cresci-2017	—	Bert	—	A: 86%
[17]	RF	UI, CF	Instagram	ROS	—	—	A: 91.76%
[18]	FFNN	UI, CF	Cresci-2017	—	GloVe, Elmo,	—	A: 94%
[19]	CNN & LSTM-Attention	CF	Cresci-2017	SMOTE	Bert	—	A: 97%
[20]	GNN, GCN	CF, NF	Cresci-rtbust, botometer-feedback, Gilani, Cresci-stock-2018, Midterm	ROS	Bert	—	A: 80%
[27]	Random walk	NF	Cresci-2015 and Social-spammer	SMOTE	Network embedding	—	Micro-F1: 97.82% Macro-F1: 97.66% AUC: 97%
[28]	DNN and LSTM	CF, UI	Midterm-18, Cresci-Rtbust and Gilani-17	—	GloVe	—	
[29]	Bert transformer	CF	Twitter dataset	—	—	—	F1-score: 97%
[30]	BiLSTM	CF, UI	Twitter dataset	—	—	—	R: 99%
[31]	BiGRU	CF, UI	Cresci-2015	ROS	GloVe	—	A: 99.44%, P: 99.25%

Bot2Vec [27] tackled the challenge of bot detection using user network features and capturing intricate local connections and node positions within the network. The experiment was conducted using the Cresci-2015 and social spammer datasets. As a result, the evaluations of Bot2Vec demonstrate superiority in efficiency and scalability compared to the current state-of-the-art techniques, with a Micro-F1 score of 97.82% and a Macro-F1 score of 97.66%, the primary limit of this research work is it's not suitable for a large number of walks per node as well as large value of walk length. DeeProBot [28] is a comprehensive deep-learning framework for Twitter bot detection based on UI and CF features extracted from the Midterm-18, Cresci-Rtbust, and Gilani-17 Datasets. Using GloVe word embedding and LSTMs, the models achieved an AUC rate of 97%. However, it cannot be generalized to all SMBs detected. This research [29] explored three classification models, XGBoost, LSTM, and transformer BERT, to detect vaccine misinformation during the COVID-19 pandemic. Trained on a dataset of over 15,000 manually annotated tweets, the BERT-based model demonstrated the best result with an F1-score of 98%. While

this method leveraged BERT's capabilities for extracting semantic meaning from textual data, its current focus on misinformation identification restricts its potential applicability to detect other types of SMBs. The authors of this article [30] proposed a system for social bot detection based on the BiLSTM model. Using features like tweet content and user metadata, the model achieved a performance of 97% across key metrics (accuracy, precision, F1-score) and 99% for recall. Even with real-world data complexities, this model demonstrated an accurate SMB identification. This work [31] presented fake account detection based on the BiLSTM model using the Cresci-15 dataset [32], including tweet content and user metadata. The results showed an accuracy of 99.44% and a precision of 99.25%. The primary limit of this approach is that it can only detect fake accounts, which restricts its potential applicability to detect other types of SMBs.

B. COMPARATIVE STUDY AND ANALYSIS

This section presents a comparative analysis of the state-of-the-art social media account classification approaches. The

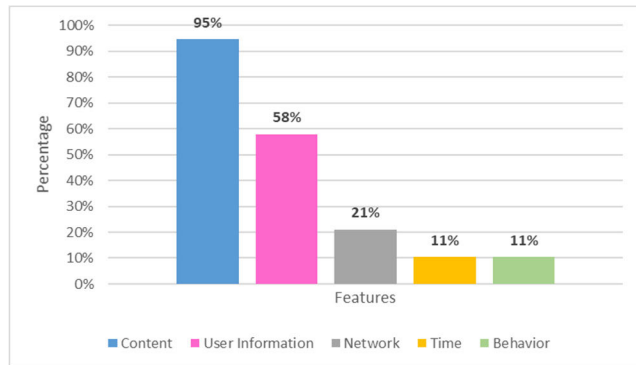


FIGURE 1. Percentage of features used for social media bots' detection.

aim is to evaluate the techniques employed by researchers in this field critically. To facilitate this analysis, we established the following set of key criteria:

- **Reference:** The references of the reviewed articles.
- **Algorithm:** The classification algorithms employed in the study.
- **Features:** Characteristics extracted from social media accounts for classification purposes, for example, User Information (UI), Content Features (CF), Network Features (NF), Behavioral Features (BF), and Timing Features (TF).
- **Dataset:** The data collection used to train and evaluate the classification model.
- **Balancing:** the balancing technique used in the article.
- **Word Embedding:** The technique employed to represent text data as numerical vectors (e.g., Word2Vec, GloVe).
- **Hyperparameters Tuning:** the process used to choose the optimal set of hyperparameters for the learning algorithm.
- **Results:** Each model's performance was measured using established evaluation metrics: Accuracy (A), Precision (P), and Recall (R).

By systematically examining these criteria, we can comprehensively understand the strengths and weaknesses of the current social media account classification approaches. This comparative analysis can guide the development of effective classification techniques for SMB detection. Table 1 presents a comparison of the state-of-the-art methods.

By analyzing Table 1, it's seen that diverse data sources were used in SMB detection models across Twitter (82%), Sina Weibo (12%), and Instagram (6%). In terms of features, content and user information features are among the most often utilized attributes for SMB detection. Figure 1 shows the percentage of the types of features used in the state-of-the-art methods.

From Figure 1, it is noticeable that the features most used by researchers for SMB detection are content features, with 95%, 58% for user information, 21% for network features, and 11% for time and behavioral features. A recent systematic literature review of SMB detection systems [4] demonstrated that the most effective features for SMBs are

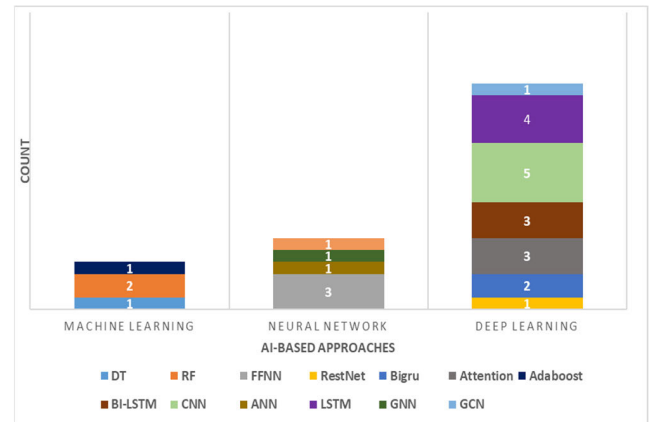


FIGURE 2. Number of AI-driven algorithms used for SMB detection.

content features and user information, reinforcing the results observed in Figure 1 and Table 1. Moreover, for the AI-driven models, the algorithms most used for SMB detection are deep learning algorithms, with a percentage of 52%, and neural network-based techniques, with a rate of 29%. The machine machine-learning-based techniques have a percentage of 19%. Figure 2 plots the statistics of the use of AI-driven techniques and algorithms in SMB detection systems.

The most commonly used algorithms for deep-learning-based approaches are LSTM and CNN. Therefore, the most commonly used neural-network-based technique is the FFNN. For machine learning algorithms, the most used algorithm is RF, and the best-achieved result is an accuracy score of 99% using the Cresci-2017 dataset. The most commonly used deep learning algorithms are CNN, LSTM, and BiLSTM.

The LSTM model achieved an F1-score of 97%, and the Bi-LSTM showed promising results with a recall of 99%. Notably, hybrid models that combine deep learning techniques, such as LSTM and CNN, surpass the unaltered approaches. These hybrid models often incorporate LSTMs, CNNs, attention mechanisms, and the BiGRU layer, achieving an accuracy of 99.44%. This suggests that combining diverse techniques and layers significantly improves the detection performance. State-of-the-art text analysis relies on word embeddings, particularly semantic embeddings, for their ability to handle both word semantics and sentence structure using vector arithmetic, which is better than numerical vector presentation. Consequently, word-embedding techniques have significantly enhanced the ability of DL models to achieve remarkable performance when handling text-based data. The most used techniques are BERT and GloVe, which improve SMB detection with an accuracy of 99.44% and a precision of 99.25%. While Table 1 shows that accuracy is the only evaluation metric employed by researchers to assess model efficacy, it's crucial to acknowledge its limitations in imbalanced classification tasks. In such scenarios, where one class significantly outnumbers the other, for example, the real users versus social media bots, a high accuracy metric can

be misleading. The model might simply prioritize predicting the majority class real users most of the time, even if its ability to identify the minority class bots is weak. Therefore, for imbalanced classification tasks, model evaluation must extend beyond solely relying on accuracy.

An examination of Table 1 reveals that the Cresci-2017 dataset was the most frequently employed dataset for social media bot detection. It comprises tweet-related information from Twitter accounts, categorized as either human or Spambot. Its publication in 2017 raises concerns about potential limitations. Social media bots have evolved significantly over time. The dataset might not fully capture the contemporary bot activity and behaviors. Furthermore, the limitation of Cresci-2017 to Spambot detection, as observed in its frequent use in training models by existing works, raises concerns about generalizability. These models might struggle to detect other types of social bots that have emerged since the dataset's creation. Therefore, it required the use of recent datasets that include different types of SMBs. The examination of Table 1 revealed several intriguing trends in the balancing approaches used in existing research works. Here is a breakdown of the main findings: 1) Half of the reviewed studies did not report using any specific balancing techniques in their methodology. This highlights the potential for addressing class imbalance to be an under-explored area in this field of research; 2) ROS emerged as the most common balancing technique, employed by 33% of the state-of-the-art works. While this approach can be effective in some cases, it can also lead to overfitting; 3) The SMOTE was utilized by 17% of the studies. SMOTE is a more sophisticated approach that addresses class imbalance by generating synthetic data points for the minority class. This analysis suggests a potential gap in the application of advanced balancing techniques. Table 1 indicates that only two of the reviewed studies employed grid search [12] and early stopping for hyperparameter optimization [28]. This finding suggests that hyperparameter optimization might be an underutilized practice within the existing research in this field. Table 1 highlights Cresci-2017 as a prominent dataset used in state-of-the-art social bot detection works.

In conclusion, this section describes the evolving techniques of SMB detection, highlighting the dominance of content features, deep learning algorithms, hybrid models, and semantic word embeddings in high-performance systems. In addition, the revelation of a large range of limitations was found by examining each of the state-of-the-art works, which raised concerns about generalizability, overfitting, and optimization.

III. BACKGROUND OF THE APPROACH

This section establishes a theoretical foundation for the deep learning algorithms employed in this study for social media bot detection. It explores the LSTM and BiGRU networks and analyzes their suitability for processing sequential data such as social media content. Subsequently, it delves into the concept of pretrained word-embedding models, specifically focusing on the role of the GloVe model in text representation.

A. LONG SHORT-TERM MEMORY

LSTM networks are variants of recurrent neural network (RNN) architecture designed to address the challenge of vanishing gradients in long sequences in traditional RNNs. To solve this problem, LSTMs combine gating and memory cells, enabling the network to preserve the data pertinent to long-range relationships. LSTM incorporates time dependency to increase the capabilities of neural network topologies. It simultaneously includes both slow- and fast-moving phenomena, and LSTM is convenient for indoor temperature modeling. It also performs well in terms of learning short- and long-term dependencies. The LSTM network structure requires four components.

- **Cell State:** The memory component of the LSTM unit that stores information for long periods of time to evade the vanishing gradients in long sequences.
- **Forget Gate:** Decides what information from the previous cell state to forget.
- **Input Gate:** Controls new information from current input.
- **Output Gate:** Decides the information from the current cell state to output as a hidden state. The value of the output Gate falls between 0 and 1.

The cells are the main LSTM units responsible for information processing. Multiple gates are incorporated into each cell to regulate and control information flow during arbitrary-length sequences. An LSTM cell, as defined in [33], can be described as follows.

$$\begin{aligned}
 i_t &= \sigma(W_{ii}x_t + b_{ii} + W_{ih}h_{t-1} + b_{hi}) \\
 f_t &= \sigma(W_{if}x_t + b_{if} + W_{ih}h_{t-1} + b_{hf}) \\
 g_t &= \tanh(W_{ig}x_t + b_{ig} + W_{hg}h_{t-1} + b_{hg}) \\
 o_t &= \sigma(W_{io}x_t + b_{io} + W_{ho}h_{t-1} + b_{ho}) \\
 c_t &= f \odot c_{t-1} + i_t \odot g_t \\
 h_t &= o_t \odot \tanh(c_t)
 \end{aligned}$$

where x and h denote the input and hidden states, respectively. The current time step is denoted by t . σ is a sigmoid activation function, and $\odot$ is the element-wise multiplication operation. g_t , f_t , i_t and o_t are the cell, forget, input, and output gates, respectively, in the current state. The learnable weights and bias terms between gates x and y are represented as W_{xy} and b_{xy} . To understand how information flows inside an LSTM cell, Figure 3. Illustration of the LSTM cell. LSTM cells can be layered as a layer and linked to one another to transfer temporal information.

B. BIDIRECTIONAL GATED RECURRENT UNIT

bidirectional gated recurrent unit (BiGRU) is a type of RNN designed to address some of the limitations of traditional RNNs. GRUs have a more straightforward internal structure than LSTM networks but still effectively capture long-term dependencies in sequential data. Unlike standard RNNs that process information only in the forward direction, BiGRUs leverage two separate GRUs: Forward GRU, which

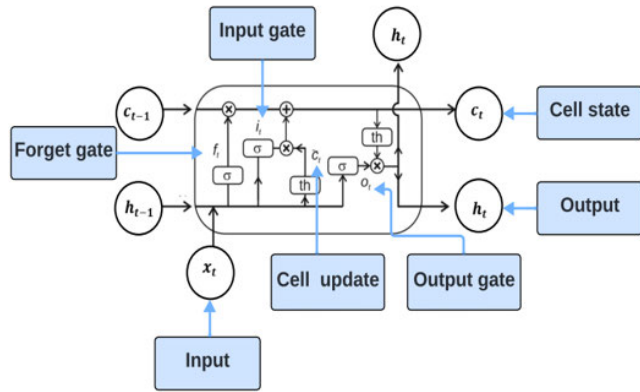


FIGURE 3. The structure of an LSTM cell.

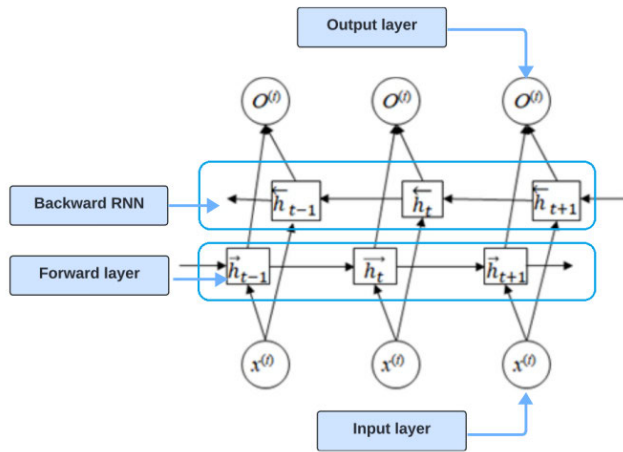


FIGURE 4. The structure of a BiGRU unit.

analyzes the sequence from beginning to end, capturing the past context of text features. The Backward GRU processes the sequence in reverse, incorporating future context. This bidirectional processing capability allows the BiGRU to learn long-range dependencies and capture the relationships between distant elements in the sequence. With a deeper contextual understanding, BiGRU can extract richer meanings from the data compared to unidirectional models by considering both past and future contexts. Figure 4 presents the internal structure of the BiGRU.

As shown in Figure 4, the BiGRU architecture comprises a forward- and backward-gated recurrent unit linked to the same output layer. In a GRU, specifically, two gates exist, known as the update gate and the reset gate, which contribute to reducing network parameters and facilitating faster convergence. The structure of the BiGRU network is illustrated in Figure 5. Then, the mathematical formulas are described.

$$\begin{aligned} r_t &= \sigma(W_r[h_{t-1}, x_t]) \\ u_t &= \sigma(W_u[h_{t-1}, x_t]) \\ \tilde{h}_t &= \varnothing(W_{\tilde{h}}[r_t * h_{t-1}, x_t]) \\ h_t &= (I - u_t) * h_{t-1} + u_t * \tilde{h}_t \\ y_t &= \sigma(W_0 * h_t) \end{aligned}$$

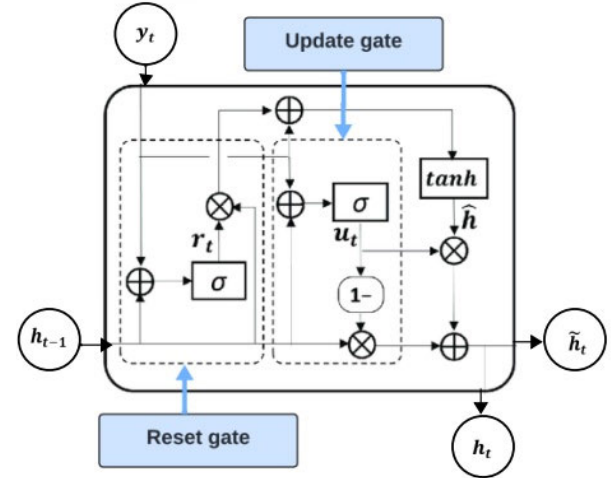


FIGURE 5. The structure of a BiGRU network.

When the input and state memory at the previous moment are denoted x_t and h_{t-1} , h_t is the state memory, which is the state of the update gate. Therefore, $\tilde{h}_t$ is the state of the current variable at the present instant, r_t the state of reset gate, u_t is the candidate set, and y_t the output vector at current moment. The corresponding inputs of the network activation functions are represented by the weight matrices $W_r, W_u, W_{\tilde{h}}, W_0$. I represents the identity matrix. $[]$ indicates the vector connection, represents the matrix dot product, and $*$ is the matrix cross product. The symbols σ and $\varnothing$ stand for the sigmoid and tanh activation functions, respectively. The following is the mathematical definition of σ and $\varnothing$:

$$\begin{aligned} \sigma(x) &= \frac{1}{1 + e^{-x}} \\ \varnothing(x) &= \frac{e^x - e^{-x}}{e^x + e^{-x}} \end{aligned}$$

The update and reset gates are the central components of the BiGRU network. The splicing matrix of the input variable x_t and the state memory variable h_{t-1} , is fed into the update gate following a sigmoid nonlinear transformation, which establishes the extent to which the state variable was brought into the current state at the previous time. The amount of data written to the candidate set is managed by the reset gate. The information of the previous time is stored by $I - u_t$ times h_{t-1} , whereas $\tilde{h}_t$ records the information from the present time. The sum of the two is the current-time output, and the sum is the current-time output.

C. GLOBAL VECTORS (GloVe)

Word embedding refers to the semantic vector space that represents a word as a numerical vector. The widespread adoption of word embedding in natural language modeling (NLP) has been attributed to its ability to encompass semantic and syntactic contexts through vector arithmetic. Global Vectors or GloVe, designed for unsupervised learning of word representations, have demonstrated superior performance in word analogy and similarity compared with other models. Its

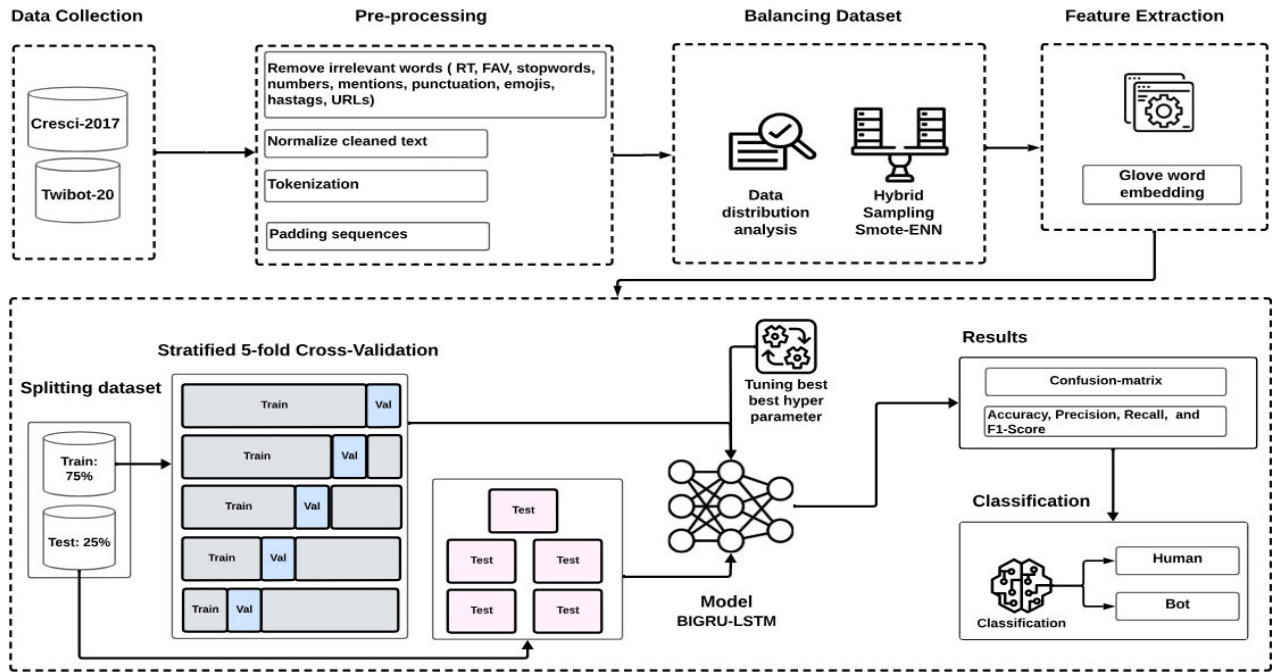


FIGURE 6. The methodology of the proposed social media bot detection system.

effectiveness surpasses that of Word2Vec owing to its capacity to capture both global and local co-occurrence counts within a corpus. This is in contrast to Word2Vec, which relies solely on local context. GloVe facilitates the creation of tweet vectors by concatenating individual word vectors within a tweet, resulting in an $n \times d$ tweet matrix, where d represents the dimension of the word vector and n denotes the tweet length [31]. GloVe has shown good results in the detection of fake accounts in this research work [31], with an accuracy of 99.44% and a precision of 99.25%. Further exploration of embeddings and integration with social media bot content features holds promise for more robust bot detection systems.

IV. METHODOLOGY

This section presents the suggested methodology for identifying social media bots. Figure 6 illustrates the detection process, which begins with gathering the datasets for social media detection. Two datasets were used for this study:

Cresci-2017, which showed promising results in state-of-the-art works, and Twibot-20, a novel dataset proposed by researchers for SMB detection. This system uses only content features. Then, all tweets posted by each user are gathered and grouped by the user in a single document. The data for each dataset were pre-processed to remove unnecessary information. Text-based features were transformed into vectors using GloVe word embedding. Subsequently, the models were trained using a hybrid architecture of the BiGRU-LSTM.

The proposed architecture for social media bot detection uses a combination of GloVe embedding, BiGRU, and LSTM

networks, which make it a unique and novel contribution to the field of social bot detection. GloVe embeddings capture semantic relationships between words, allowing for a better understanding of meaning and intent in social media text-based posts. Long-range contextual information is captured by BiGRUs, while the sequential nature of social media text is handled by LSTMs. This combined approach allows the model to learn more intricate representations of social media posts, potentially leading to more accurate identification of bots that mimic human language patterns, which offers several advantages, including enhanced semantic understanding, bidirectional dependency learning, and improved sequential pattern recognition. This architecture offers potential novelty in the context of social media bot detection, potentially improving performance compared to the existing models proposed by researchers. Figure 6 illustrates the proposed methodology.

As shown in Figure 6, the architecture of the proposed system includes different stages: 1) dataset collection, 2) data pre-processing, 3) balancing the datasets, 4) feature extraction, and 5) building the model. A comprehensive explanation of the five constituent stages of the proposed SMB detection system is presented in this section.

A. DATASET COLLECTION

By analyzing state-of-the-art research, we found that the Cresci-2017 dataset is the most frequently used by researchers according to assessments of cutting-edge research. The “Cresci-2017” [34] dataset is a widely used dataset in the field of social media bot detection. It consists of

TABLE 2. Datasets description.

Dataset name	Content	Description	Size
Twibot-20	D1: Twitter dataset	Dataset of the recent generation of Twitter bots and genuine users.	Users: 229573, tweets: 33,48m, followers: 455958, year: 2020
Cresci-17	D1: genuine accounts	Human accounts	Users: 3474, tweets: 8.377, year: 2011
	D2: Social spam bots1	Retweeters of the political candidate from Italy	Users: 991, tweets: 1.61m, year: 2012
	D3: Social spam bots2	Spammers of paid mobile applications	Users: 3457, tweets: 428542, year: 2014
	D4: Social spam bots3	Amazon.com spammers	Users: 464, tweets: 1.41m, year: 2011
	D5: Traditional spam bots1	Spammers	Users: 1000, tweets: 145094, year: 2009
	D6: Traditional spam bots2	Spammers using scam URLs	Users: 100, tweets: 74957, year: 2014
	D6: Traditional spam bots3	Spammers of job offer	Users: 433, tweets: 5.79m, year: 2013
	D7: Traditional spam bots4	Spammers of job offers	Users: 1128, tweets: 133311, year: 2009
	D8: Fake followers	Accounts of fake followers	Users: 3351, tweets: 196027, year: 2012

Twitter accounts labeled as either genuine or bots. Accounts were collected using various strategies, including the following known bots and gathering accounts through publicly available lists. The data included metadata, tweet content, and social network characteristics. However, the Cresci-2017 dataset has potential biases due to its data collection methods, data aging, and limited spambot detection. On the other hand, the “Twibot-20” dataset is a recent dataset related to bot detection on Twitter. It contains information about both real users and SMBs, making it a comprehensive dataset for evaluating bot-detection models. It includes information about approximately 1.5 million Twitter users collected in 2020, including user profile information (e.g., bio, follower/following count, tweet activity), network features, etc. To train the proposed models, the experiment leveraged two datasets, Cresci-2017 and Twibot-20. A detailed description of these datasets is provided in Table 2.

B. DATA PRE-PROCESSING

The OSN-based data include user-generated content, including tweets. However, these data contain noise and inconsistencies that can hinder the effectiveness of the SMB detection models. A crucial step involves data pre-processing to ensure optimal model performance. This section describes the pre-processing techniques employed to prepare a clean input for the proposed system.

1) CLEANING DATA

Text pre-processing is a fundamental stage in Natural Language Processing (NLP) pipelines. The data cleaning pipeline involves the elimination of HTML hashtags, URLs, mentions, punctuation marks, numerical digits, emojis, word elongations, single letters, stop words, repeated letters, and diacritics. Eventually, tokens were normalized to their initial states and transformed to miniscule.

2) TOKENIZATION AND PADDING

Tokenization and padding are two fundamental pre-processing techniques. Tokenization involves splitting text into smaller units called tokens. This step converts raw text

data into a format that machine-learning models can process. In padding sequences in deep-learning models, inputs often require fixed dimensions. However, text sequences can vary in their length. Padding adds dummy tokens to sequences shorter than a specified length, making all sequences in the dataset equal in size. This ensures that the inputs to the model have consistent dimensions, which are necessary for an efficient computation. After tokenization and padding, text data can be fed into the deep learning models.

C. BALANCING THE DATASETS AND DATA PREPARATION

Addressing class imbalance in datasets is a crucial step in ensuring the robustness of the detection model. A primary concern when addressing class imbalance is the potential for overfitting, especially with oversampling techniques. To get rid of overfitting SMOTE (Synthetic Minority Oversampling Technique) and its advanced variant, SMOTE-ENN (Edited Nearest Neighbors) were employed. Unlike random oversampling, which duplicates existing minority class data points, SMOTE generates synthetic samples, increasing the size and diversity of the minority class while mitigating overfitting risks. However, the original SMOTE has limitations in handling complex data and borderline cases were minority class points near the majority class. SMOTE-ENN addresses these issues by incorporating Edited Nearest Neighbors, potentially leading to better model generalization. Considering the imbalanced distribution, Figure 7 plots the imbalanced distribution of datasets, particularly in Cresci-2017.

D. FEATURE EXTRACTION

Feature extraction is a crucial step in NLP with the aim of transforming text-based data into numerical representations suitable for ML and DL models. This process involves identifying significant textual patterns and attributes that capture the semantic meaning and context of the text in a format that can be understood by a classifier. This study leverages GloVe word embedding for the text presentation of the training data. In feature extraction with GloVe, each word in the text is represented by a fixed-length vector derived from pre-trained GloVe embeddings. These vectors encode

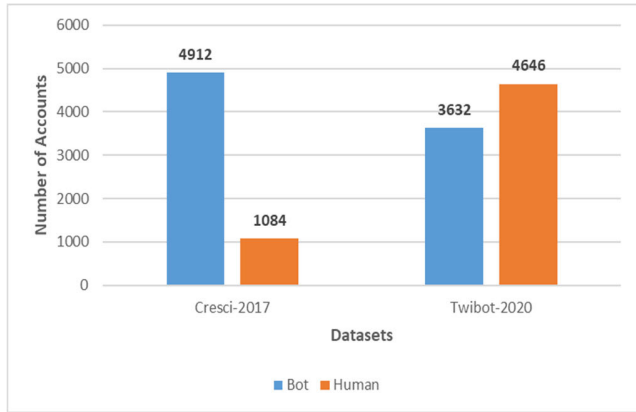


FIGURE 7. Datasets distribution.

semantic information regarding the meanings and contexts of the words. This process results in a fixed-length vector representation of text. Overall, feature extraction using GloVe involves leveraging pretrained word embeddings to represent text data in a numerical format, enabling the use of the data as input for the proposed SMBs detection system.

E. BUILDING THE MODEL

After preparing the data for the training of the model, the dataset was partitioned into training and testing sets, allocating a split ratio of 25% for testing and 75% for the training set. GloVe is not inherently adept at capturing bidirectional textual context. To address this limitation, this system combines the strengths of GloVe with the sequential and bidirectional capabilities of the BiGRUs and LSTM networks. To address the class imbalance and model evaluation in imbalanced classification tasks, the following strategies were employed to mitigate the challenges associated with imbalanced datasets and ensure a robust model. 1) A Multifaceted Evaluation Approach: While accuracy remains a fundamental metric for model performance, its shortcomings become evident when dealing with imbalanced data. To achieve a more comprehensive evaluation, we employed additional performance measures alongside accuracy. These include precision, recall, and F1-score. 2) Stratified K-Fold Cross-Validation for Robust Evaluation: to assess the model's generalizability and overcome the challenges associated with imbalanced datasets, stratified 5-fold cross-validation was employed. This technique generates multiple, unique training and testing sets and offers a significant advantage over traditional K-fold cross-validation in imbalanced classification tasks. Stratified K-fold cross-validation ensures that each fold maintains a class distribution mirroring the overall dataset. This characteristic is particularly crucial when the minority class holds significant importance for the model's efficacy, as in our case with the Cresci-2017 dataset, we investigated the performance of Stratified K-Fold Cross-Validation across a range of K values [3], [5], [10], the K=5 provided the best results and computational efficiency. 3) Hyperparameter Optimization: This paper examines two techniques for hyperparameter

TABLE 3. Results of hyperparameters tuning.

Hyperparameters	Value
Max_vocabulary	4000
Input_max_length	50
Embedding_size	100
Learning_rate	0.001
Batch_size	64
Epochs	100
Optimizer	Adam
Stratified K-Fold Cross-Validation	K=5

tuning, Grid Search and Random Search to find the optimal Hyperparameter, table 3 provides the list of hyperparameters values. 4) Early stopping is employed to mitigate the risk of overfitting during the model training process. This approach entails monitoring the model's performance on a validation set throughout training. If the performance on the validation set fails to improve for a predefined number of epochs, the training process is stopped. This strategy effectively prevents the model from memorizing the training data excessively, leading to improved generalizability of unseen data. 5) Generalizability and Model Evaluation: evaluating the model's generalizability and its ability to perform well on unseen data is a cornerstone of the training process. This evaluation is typically achieved through testing the model on a dedicated dataset that has not been used during training. High performance on this test set signifies the model's capacity to learn generalizable patterns that are applicable to unseen data, ultimately reflecting its potential effectiveness in real-world scenarios; 6) Early Stopping to Enhance Training Efficiency: Early stopping is employed to mitigate the risk of overfitting during the model training process. If the performance on the validation set fails to improve for a predefined number of epochs, the training process is stopped. This strategy effectively prevents the model from memorizing the training data excessively, leading to improved generalizability of unseen data. Figure 8 shows the architecture of the proposed model.

Similar to other deep learning neural networks, the proposed system is a multi-layered architecture that combines information processing and data learning. The layers comprising the proposed hybrid RNN architecture are illustrated in Figure 8 and described as follows:

- **Input layer:**The input layer is the initial stage of a model that receives and processes data. It accepts cleaned datasets, ensuring quality and suitability. The model aims to categorize data into two distinct classes: SMBs and "human accounts". The data is labeled according to these categories. The input layer is configured to handle sequences of text data with a maximum vocabulary size of 4000, a maximum length of 50 tokens, and an embedding size of 100.
- **Embedding Layer:** This layer maps individual words from a predefined vocabulary of size vocabulary_size into dense numerical vectors of dimension

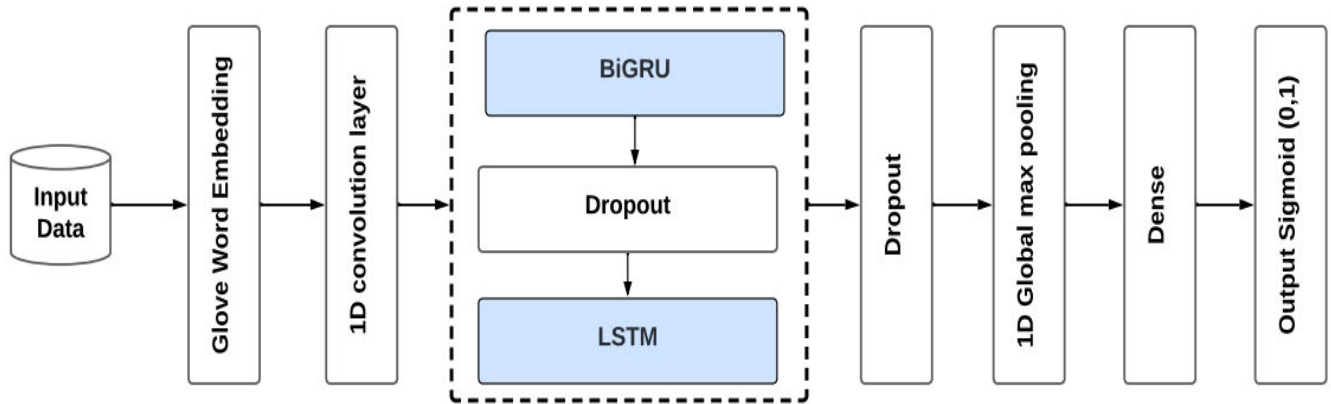


FIGURE 8. The architecture of the proposed methodology.

embedding_dim. These embedding vectors aim to capture the semantic relationships between words, allowing the model to represent words based on their meaning and context. Notably, the vocabulary size is set equal to the maximum vocabulary size max_vocabulary, and the embedding dimension embedding_dim is set to 100.

- **Conv1D Layer:** This layer captures local features in the sequence, helping to identify patterns relevant to understanding the data.
- **BiGRU and LSTM Layers:** RNN layers can learn and remember long-range dependencies in data. The bidirectional nature of the GRU allows the model to capture information from past and future contexts.
- **Dropout Layers:** prevent overfitting by introducing noise during training, reduce reliance on specific neurons, and encourage the model to be more robust.
- **GlobalMaxPooling1D Layer:** Extracts the most crucial feature from the sequence, providing a global view of the information captured by the previous layers.
- **Dense Layer:** This layer takes the output of the previous layers and produces the final classification result using a sigmoid activation function. The model learns to make predictions based on the hierarchical features extracted from the input sequence. Because it is a binary classification problem, a sigmoid function is used for the output layer.

Table 4 lists the model layers and hyperparameters. This will help future readers reproduce the experimentation in their research.

The model was trained and tested using a 5-fold stratified cross-validation to ensure reliable performance and avoid overfitting. This method splits the data into smaller groups, uses each group for testing at different times, and trains the remaining groups.

V. RESULTS AND DISCUSSION

This section evaluates the proposed BiGRU-LSTM architecture for social media bot detection. Experiments were

TABLE 4. The parameters of the model layers.

Layer	Output Shape	Parameters
InputLayer	[(None, 50)]	max_length= 50
Embedding_layer	(None, 50, 100)	input_dim= 4000, output_dim= 100, weights=[embedding_matrix], input_length=50, trainable=False
Conv1D	(None, 46, 128)	filters=128, kernel_size=5, activation='relu'
BiGRU_layer1	(None, 46, 200)	Bidirectional (GRU (100, return_sequences=True))
Dropout	(None, 46, 200)	Dropout (0.5)
LSTM	(None, 46, 100)	(100, return_sequences=True)
Dropout	(None, 46, 100)	Dropout (0.5)
GlobalMaxPooling1D	(None, 100)	—
Dense	(None,1)	—

conducted using two benchmark datasets: Cresci-2017 and Twibot-20. The following subsections highlight these findings and demonstrate the efficacy of the methodology for classifying SMBs. This section presents and evaluates the results obtained using different techniques and metrics. The model was trained on Google Colab Pro with 108.26 calculation units, 40GB RAM, and 225.8 GB of disk memory. The main models used to train the model were Keras (2.2.4), TensorFlow (2.12.0), Python (3.10.12), and Jupyter (6.1.12).

A. PERFORMANCE METRICS

Several metrics were employed to test, validate, and assess the model's efficiency in identifying SMBs, including accuracy, precision, recall, F1-score, and AUC. The true positive (TP) signifies a correctly classified SMB, whereas a false positive (FP) indicates a legitimate account incorrectly classified as

a bot. Conversely, a true negative (TN) represents a correctly identified legitimate account, and a false negative (FN) denotes a missed SMB. The definitions of these measures are provided below.

1) ACCURACY

measures the proportion of correctly predicted or classified instances out of the total number of instances in a dataset. It is calculated as:

$$\text{Accuracy} = \frac{\text{Number of Correct Predictions}}{\text{Total Number of Predictions}}$$

2) RECALL

also known as Sensitivity or True Positive Rate, measures the ability of a classification model to correctly identify all relevant instances or true positives (TP) within a dataset. Recall is computed using the following formula:

$$\text{Recall} = \frac{\text{Number of TP}}{\text{Number of TP} + \text{Number of FN}}$$

3) PRECISION

referred to as specificity in some contexts, is a machine learning metric that measures the accuracy of positive predictions made by a classification model and is computed by the following formula:

$$\text{Precision} = \frac{\text{Number of TP}}{\text{Number of TP} + \text{Number of FP}}$$

4) F1 SCORE

This is an evaluation metric that combines precision and recall into a single value to provide a balanced measure of a classification model's overall performance. The F1 Score was calculated using the following formula:

$$F1 - \text{score} = \frac{2 * (\text{Precision} * \text{Recall})}{\text{Precision} + \text{Recall}}$$

5) AUC-ROC

This metric reflects the system's ability to classify bots and legitimate users correctly. It helps assess the effectiveness of the model in differentiating between real and fake users.

B. APPROACH EVALUATION AND COMPARISON

This section presents a performance evaluation of the proposed BiGRU-LSTM social media bot detection approach. The obtained results are presented and compared with the performance of other state-of-the-art techniques in this field. Table 5 presents the confusion matrix of the proposed approach using the Cresci-2017 and Twibot-20 datasets using the ROS balancing data.

It is noticeable from Table 5 that the models trained on Cresci-2017 achieved higher results compared with the model trained on the Twibot-20 dataset with a TP rate equal to 100%, TN rate of 100%, FP rate of 0%, and FN rate of 0%. However, the model trained on the Twibot-20 achieved a TP rate equal to 99.67%, a TN rate of 99.66%, an FP rate of

TABLE 5. Confusion matrix of the trained models.

Dataset	TP	TN	FP	FN
Twibot-20	99.67%	99.66%	0.04%	0.03
Cresci-2017	100%	100%	0%	0%

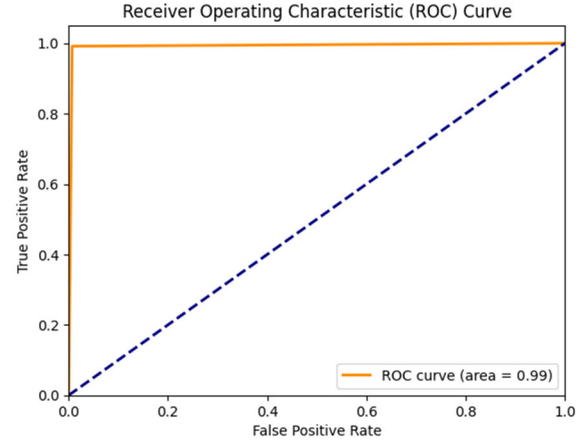


FIGURE 9. ROC curve of the models trained on the Twibot-20 dataset.

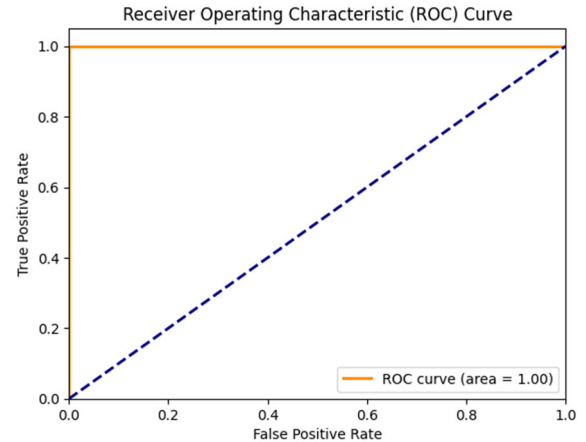


FIGURE 10. ROC curve of the models trained on the Cresci-2017 dataset.

0.04%, and an FN rate of 0.03%. To evaluate the trade-off between sensitivity (TP) and specificity (TN). In addition, Figures 9 and 10 plot the Receiver Operating Characteristic (ROC) curve based on the sensitivity and specificity rates.

It's observable from Figure 9 and Figure 10 that for Twibot-20, the model reached a score of 99%, and for Cresci-2017, the model reached a score of 100%. Such results often indicate overfitting, especially when the performance metrics are perfect. To address concerns about the potential for overfitting due to ROS, SMOTE and SMOTE-ENN are employed as a balancing technique to mitigate overfitting issues. Unlike ROS, which duplicates existing minority class data points randomly, SMOTE creates synthetic data points to increase the size and diversity of the minority class while mitigating the risk of overfitting. However, the original SMOTE has limitations, including data complexity and borderline

TABLE 6. Test results of the trained models using cresci-2017 and Twibot-20.

Dataset	Approaches	Precision	Accuracy	Recall	F1-Score
Twibot-20	GloVe, ROS, BIGRU, LSTM	99.66%	99.61%	99.66%	99.66%
	GloVe, SMOTE, BIGRU, LSTM	99.45%	98.12	96.77%	98.09%
	GloVe, SMOTE-ENN, BIGRU, LSTM	100%	99.73%	99.56%	99.63%
Cresci-2017	GloVe, ROS, BIGRU, LSTM	100%	100%	100%	100%
	GloVe, SMOTE, BIGRU, LSTM	99.91%	99.92%	99.91%	99.91%
	GloVe, SMOTE-ENN, BIGRU, LSTM	93.18%	98.11%	89.13%	91.11%

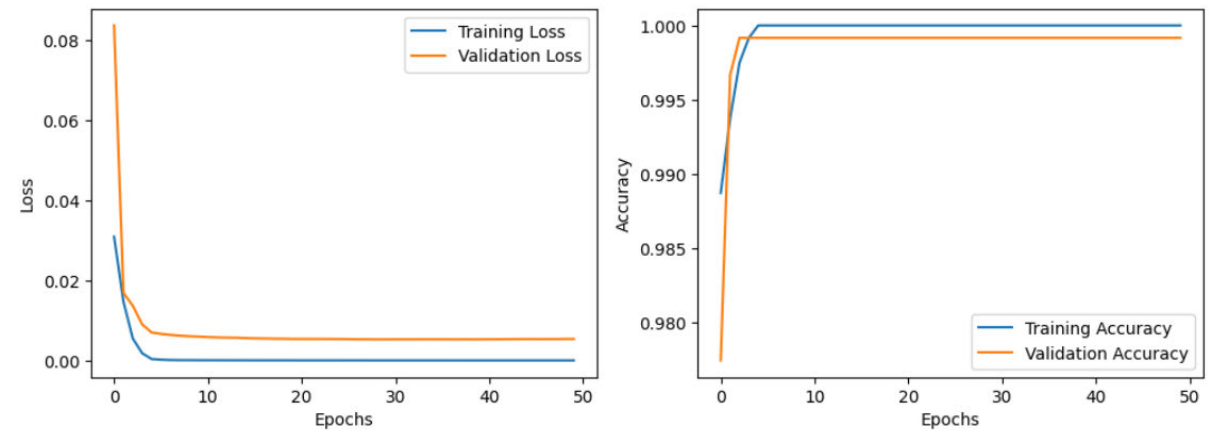


FIGURE 11. Cross-validation results using the Twibot-20 dataset.

TABLE 7. The model test results on unseen data.

Data size	Results
Human: 983	Precision: 98.89%, Recall: 91.35%, F1-Score: 94.97%, and Accuracy:92.08%
Bot: 217	

cases, as well as data points from the minority class that lie close to the majority class. Researchers have developed variants to address these issues. The SMOTE-ENN advanced SMOTE variations that tackle imbalanced data by combining over- and under-sampling using SMOTE and Edited Nearest Neighbors, potentially leading to better generalization. Table 6 provides a detailed presentation of the performance of the trained BiGRU-LSTM models on the Cresci-2017 and Twibot-20 datasets. The evaluation employed three distinct balancing techniques: ROS, SMOTE, and SMOTE-ENN. This table offers a comprehensive overview by presenting a range of evaluation metrics, including precision, accuracy, recall, and F1-score. By analyzing these metrics across both datasets, we can examine the effectiveness of the BiGRU-LSTM architecture for SMB detection.

Table 6 presents the results of the BiGRU-LSTM model, trained on two SMB detection datasets, Cresci-2017 and Twibot-20, using three different balancing techniques: ROS,

SMOTE, and SMOTE-ENN. All the trained models demonstrated high performance across all balancing techniques with Precision, Recall, F1-Score, and Accuracy exceeding 96%. SMOTE-ENN for Twibot-20 achieved the highest values in all metrics with a precision of 100%, accuracy of 99.73%, recall of 99.56%, and F1-Score of 99.63%. The evaluation of the results suggests that combining GloVe embeddings, SMOTE-ENN for class imbalance mitigation, and a BiGRU-LSTM architecture yielded the most promising results for SMB detection using the Twibot-20 dataset. However, it is crucial to acknowledge limitations. The Cresci-2017 dataset might not fully capture the evolving landscape of social bots type due to its limitation to Spambots, potentially hindering the model’s generalizability to detect diverse bot behaviors in the real world. Conversely, the strong performance of the model on Twibot-20 can be attributed to its recent nature, encompassing a more comprehensive range of social media content types, and the effectiveness of SMOTE-ENN in addressing class imbalance without introducing significant bias or overfitting. To validate the models and ensure reliable performance that avoids overfitting, cross-validation was used with a stratified 5-fold cross-validation. Figure 11 provides the cross-validation curves of the training loss and accuracy of the model trained on the Twibot-20 datasets. The training loss reflects how well the model fits the training set data. The validation loss is the error a model

TABLE 8. Comparison of BiGRU-LSTM model performance with the state-of-the-art approaches.

Approaches	Baseline models	Dataset	Precision	Accuracy	Recall	F1-Score
BiGRU-LSTM	GloVe, SMOTE-ENN, BIGRU, LSTM	Twibot-20	100%	99.73%	99.56%	99.63%
	GloVe, SMOTE, BIGRU, LSTM	Cresci-2017	99.91%	99.92%	99.91%	99.91%
[12]	BERT, FFNN	Cresci-2017	—	95%	—	—
[8]	RF	Cresci-2017	—	99%	—	—
[14]	GloVe, BiLSTM	Cresci-2017	—	92.9%	—	—
[15]	Bert, CNN, ANN	Cresci-2017	—	>90%	—	—
[16]	Bert, FFNN	Cresci-2017	—	86%	—	—
[18]	GloVe, Elmo, FFNN	Cresci-2017	—	94%	—	—
[19]	Bert, CNN & LSTM, Attention layer	Cresci-2017	—	97%	—	—

has on data that is not trained on. This indicates how well the model generalizes and performs well on unseen data.

Figure 11 shows that for Twibot-20, the scores reached were 0.01 for the training loss and 0.02 for validation. The results presented in this section demonstrate the effectiveness of the proposed social media bot detection system. A crucial aspect of the training process involves assessing the model's generalizability and ability to perform effectively on unseen data. This evaluation is commonly achieved by employing a dedicated test set that remains separate from the training data. Strong performance on this unseen data suggests the model's capacity to capture generalizable patterns applicable beyond the training examples. This, in turn, reflects its potential effectiveness for real-world deployments. Table 7 presents the results of testing the model on unseen OSN data. The dataset includes text-based content of different types of SMBs collected from Twitter in 2022 [35], and the dataset is dedicated to graph-based social media bots detection.

The BiGRU-LSTM model's performance on unseen data from a 2022 social media platform showed promising results, with a precision of 98.89% and an F1-score of 94.97%. The model performed well in identifying humans and bots, but its accuracy dropped from 99.73% to 92.08% on unseen data, indicating challenges in generalizability. The model's recall also decreased slightly from 98.86% to 91.35%, suggesting it might have missed some SMBs in detection. The difficulty level of the unseen data is unknown, so further analysis with a more extensive and diverse dataset is recommended. The initial evaluation of unseen data is encouraging. Still, additional study with a more extensive and diverse dataset is recommended for a more comprehensive understanding of the model's generalizability and potential limitations in real-world social bot detection scenarios. To compare different SMB detection approaches, it is crucial to use the same dataset. Various datasets were used for the reviewed articles. Moreover, not all approaches necessarily include different values of the performance metrics, which limits the comparison. Therefore, the proposed system must be thoroughly

compared with existing works. The comparison included only reviewed articles that used the Cresci-2017 dataset, which showed the highest performance. To confirm the effectiveness of the GloVe word embedding and BiGRU-LSTM model, Table 8 presents a comparative analysis of the performance of BiGRU-LSTM against state-of-the-art approaches. This table facilitates the evaluation of the efficacy of the proposed methodology relative to the existing systems in the social media bot detection field.

As indicated in Table 8, the best results were obtained with the BiGRU-LSTM model trained on Twibot-20, which reached the highest values in all metrics with a precision of 100%, accuracy of 99.73%, recall of 99.56%, and F1-Score of 99.63%. However, for the Cresci-2017, the achieved scores were 99.91% for accuracy and a precision of 99.91%. On the other hand, the best performance shown in the state-of-the-art techniques is the accuracy of 97% presented in this article [19], with a model implemented using Bert, CNN, LSTM, and an attention layer, and trained on the Cresci-2017 dataset. This comparative analysis highlights the proposed method's effectiveness compared to the existing methodologies.

C. DISCUSSION

The BiGRU-LSTM architecture has the potential to be an effective tool for social media bot detection because it significantly outperforms previously published models. This architecture leverages a hybrid approach that combines these two key strengths. First, using GloVe Word Embedding, this technique captures the semantic meaning of a text by leveraging a large vocabulary. This allows training data to preserve a wider range of word associations, crucial for text presentation and analysis of OSN-based content features. Second is the hybridization of BiGRU and LSTMs. BiGRU facilitates context extraction from text data by analyzing sequences in both directions using forward and backward RNN. This enables the model to grasp content features within a broader context and learn word relationships. In addition, LSTMs handle sequential data, capture long-term dependencies, and

identify relevant features for SMB detection. Their ability to analyze sentence relationships, vocabulary usage, and sentence structure enhances the overall accuracy of social media bot detection and contributes to a highly effective system for SMB detection. Different factors influence the computational complexity of the BiGRU-LSTM model. Time complexity, which is the time required to train the model, depends on 1) Number of layers: The BiGRU-LSTM architecture includes nine layers, as shown in Table 4. Each layer contributes to the overall training time. Generally, more layers increase model complexity and training time; 2) Sequence Length: The hyperparameter `max_length = 50` defines the maximum sequence length processed by the network. Longer sequences require more processing steps within the BiGRU-LSTM, increasing training time. The space complexity of the BiGRU-LSTM network depends on: 1) Embedding Dimensionality: The hyperparameter `embedding_size = 100` defines the dimensionality of the word embeddings, a higher dimensionality requires more memory to store the embedding vectors; 2) The Vocabulary Size: the hyperparameter `max_vocabulary` suggests the potential maximum vocabulary size which is equal to 4000 tokens, a larger vocabulary translates to a larger embedding matrix, impacting memory usage. In addition to these elements, our system also includes a validation phase. Using Stratified 5-fold Cross-Validation introduces additional computational overhead but is typically negligible compared to the training time. Based on the actual configuration of our training environment using Google Colab Pro, which was already described at the beginning of this section, the actual computational time of 10 minutes was measured during the training process on our specific hardware configuration. Performance scaling of the model is evaluated based on accuracy, precision, and recall of unseen data and a recent dataset, which showed good results. This analysis aims to enhance transparency, identify trade-offs between model complexity, training time, and performance on unseen data, and strengthen the model's effectiveness. Despite the good results of the proposed architecture, it is crucial to acknowledge the inherent limitations of any methodology. Specifically, generalizability concerns exist. The model was trained only for text-based features. Therefore, they cannot be generalized to complex datasets that include behavioral and network features. In addition, this system is designed for the binary classification of OSN accounts (human or bot), and its applicability in detecting other types of social media bots may represent a limitation of efficient classification. In addition, detecting social media bots in different languages presents challenges such as linguistic variations, cultural nuances, and limited resources. To overcome these, potential methods include multilingual language models, transfer learning, and identifying language-agnostic features like network activity and posting frequency, hashtags, mentions, retweets, emojis, etc. These methods reduce the need for language-specific training data and enable effective bot detection across multiple languages. Future research should explore the complexities

of cross-lingual bot detection, examine language-specific challenges, and explore the effectiveness of pre-trained multilingual language models, transfer learning techniques, and language-agnostic features. Indeed, the proposed architecture can be adapted for tasks like spam detection. It might effectively identify spam messages that often exhibit repetitive language or manipulative content. Similarly, the model's capacity to analyze textual content could be valuable for tasks like fake news identification, where the ability to discern factual language from misleading narratives is crucial. Additionally, by incorporating sentiment analysis techniques alongside the proposed architecture, the model could potentially be used to analyze the emotional tone of online content, providing valuable insights for various research applications. To enhance the robustness of the model in capturing wider types of SMBs, the model has to be trained on datasets that include other types of SMBs (e.g., spam bots, Sybils, and Cyborg). This can open the axis of research to propose new solutions to these problems by exploring more advanced neural network architectures that could improve detection performance. Investigating new strategies to handle imbalanced datasets and implementing an active learning process would enable the model to adapt to incoming data streams in real-time.

VI. CONCLUSION

This study proposes a novel methodology for SMB detection based only on the content features of OSN accounts. This work combines the GloVe method for the semantic representation of tweets and a hybrid RNN architecture, including BiGRU and LSTM models. For the training step, the Cresci-2017 and Twibot-20 standard datasets were employed. The strength of this contribution relies on every stage of data mining in building the classifier model, including dataset pre-processing and cleaning, balancing datasets using the SMOTE-ENN method, GloVe for feature extraction, and training the hybrid model using stratified 5-fold cross-validation to avoid overfitting and ensure better generalization. The performance score of the new architecture combining BiGRU and LSTM models was impressive on both datasets using the Twibot-20 achieved the highest values in all metrics with a precision of 100%, accuracy of 99.73%, recall of 99.56%, and F1-Score of 99.63%. For the Cresci-2017, the model achieved the highest values in all metrics with a precision of 99.91%, accuracy of 99.92%, recall of 99.91%, and F1-Score of 99.91%. This approach demonstrated promising potential in identifying social bots by achieving performance that surpasses the threshold accuracy of 97% shown by the state-of-the-art methods. Then, the model trained on Twibot-20 was evaluated using unseen and recent data, achieving an accuracy of 92.08%. Therefore, the success of this methodology can be attributed to the combination of GloVe word embedding, which effectively captures the global co-occurrence information in an extensive range of words. Moreover, the hybrid BiGRU-LSTM architecture allows for the extraction of bidirectional and long-term

relationships within the text sequences. This approach can be explored beyond social media bot detection, such as spam detection, fake news identification, or sentiment analysis, which can open axes for future works of OSN analysis and classification. Future investigations will focus on developing new methodologies that effectively detect SMBs in different languages, such as Arabic, French, German, and Spanish. Detecting social media bots in different languages presents challenges such as linguistic variations, cultural nuances, and limited resources. To overcome these, potential methods include multilingual language models, transfer learning, and language-agnostic features. These methods can reduce the need for language-specific training data and leverage pre-trained models for multiple languages. Additionally, identifying features that are not language-dependent can help detect bots across languages. Furthermore, the model was trained only for text-based features, which cannot be generalized to complex datasets that include different types of features. For future works we propose to a multi-model system that can detect SMBs using an extensive range of attributes.

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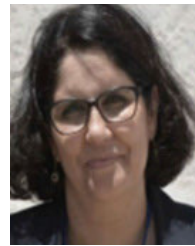
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